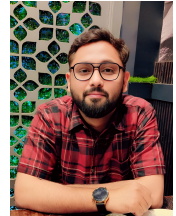


Aizan Zafar, Ph.D.

✉ aizanzafar@gmail.com
🎓 Google Scholar

in LinkedIn

🌐 Portfolio



Professional Summary

- Ph.D. in Computer Science with expertise in **vision-language models**, **causal reasoning**, and **knowledge-grounded AI**.
- Postdoctoral Researcher at **IIIT Bangalore** (with Siemens), focusing on industrial AI: PLC/SCL code modeling, neural-symbolic methods, and knowledge graphs.
- Former Postdoctoral Scholar at **CRCV, UCF**, developing multimodal medical AI for VQA, causal reasoning, and visual grounding.
- Publications in EMNLP, COLING, WACV, and top journals (Springer, Elsevier, Nature).
- Passionate about building interpretable, scalable AI systems for healthcare, industrial automation, and geolocation.

Employment History

August 2025 – Present

■ **Postdoctoral Researcher**, International Institute of Information Technology Bangalore (IIIT-B), India
Collaboration: Siemens (Industrial AI)

- Leading research on industrial AI with focus on **PLC/SCL code modeling**, **neural-symbolic methods**, and **knowledge-grounded representations**.
- Building **SCL Knowledge Graphs (SCLKG)** for automated test-case generation, impact analysis, and requirements traceability.
- Developing LLM/VLM-based tools for code summarization, anomaly detection, and HMI metadata mapping.
- Coordinating joint research milestones and deliverables with Siemens R&D teams.

Oct 2024 – August 2025

■ **Postdoctoral Scholar**, Center for Research in Computer Vision (CRCV), University of Central Florida, Orlando, USA
Advisor: Prof. Mubarak Shah

- Researched vision-language models with focus on **visual grounding**, **multimodal reasoning**, and **causal inference** in medical and geolocation domains.
- Developed **CRANE**, a causal reasoning framework for medical VQA using structured knowledge graphs and visual attention alignment (*Transactions on Medical Imaging*, under review).
- Co-developed scalable multimodal pipelines for **geolocation reasoning**, leading to **GAEA** (WACV 2026, accepted).

Education

- July 2019 – Feb 2025  **Ph.D., Computer Science and Engineering**
Indian Institute of Technology (IIT) Patna, Bihar, India
Thesis: Enhancing Medical Consultations with Knowledge-Grounded Question Answering and Dialogue Generation
Advisor: Associate Prof. Asif Ekbal
- Aug 2017 – June 2019  **M.Tech, Information Technology**
University of Hyderabad, Telangana, India
Thesis: An Efficient Approach for Effective Clustering Using Validity Index
Advisor: Prof. K. Swarupa Rani
- Aug 2012 – June 2016  **B.Tech, Information Technology**
Guru Ghasidas Central University, Chhattisgarh, India

Research Publications

Conference Proceedings

- **Aizan Zafar**, Sreekantam Sai Venkata, Gathik Jindal, Ritu Raj Sharma, Zishan Ahmad, Tulika Saha, Srinath Srinivasa "TrustMod-SM: A Multi-Axis Benchmark for Evaluating Trustworthiness of LLMs in Social Media Content Moderation" **NeurIPS 2026 Evaluations and Datasets Track**, under review, 2026.
- **Aizan Zafar**, Mayank Chamarthi, Tulika Saha, Mamta. "IndicTableQA: A Large-Scale Multilingual Benchmark for Table Question Answering in Low-Resource Indian Languages." Under review.
- Sumra khan, Sagar Chhabriya, **Aizan Zafar**, Sheeraz Arif, Amgad Muneer, Anas Zafar, Shaina Raza, Rizwan Qureshi "Towards Responsible Multimodal Medical Reasoning via Context-Aligned Vision-Language Models" Accepted at **CVPR 2026 Workshop KnowledgeMR**.
A context-aligned reasoning framework that enforces agreement across heterogeneous clinical evidence for generating diagnostic conclusions.
- Ron Campos, Ashmal Vayani, Parth Parag Kulkarni, Rohit Gupta, **Aizan Zafar**, Aritra Dutta, Mubarak Shah. "GAEA: A Geolocation Aware Conversational Assistant." In Proceedings of **WACV 2026**.
An interactive geolocalization chatbot; moving beyond raw coordinates, GAEA provides rich contextual insights about locations directly from images.
- **Aizan Zafar**, Kshitij Mishra, Asif Ekbal. "MedEx: Enhancing Medical Question Answering with First-Order Logic Based Reasoning and Knowledge Injection." In Proceedings of the **31st International Conference on Computational Linguistics (COLING 2025)**, Abu Dhabi, UAE. (**CORE Rank: A**)
Integrates First-Order Logic (FOL) reasoning with knowledge graphs to model symptom–disease–treatment relationships, enhancing medical QA accuracy.
- **Aizan Zafar**, Kshitij Mishra, Asif Ekbal. "MedLogic-AQA: Enhancing Medicare Question Answering with Abstractive Models Focusing on Logical Structures." In **Findings of ACL: EMNLP 2024**, pp. 16844–16867. (**CORE Rank: A***)
Leverages First-Order Logic rules with neural abstractive models to improve logical reasoning in medical QA, achieving state-of-the-art results.
- Dibyanayan Bandyopadhyay, **Aizan Zafar**, Asif Ekbal, Mohammed Hasanuzzaman. "End-to-End Sign Language Translation via Multitask Learning." In Proceedings of **IJCNN 2023**.
Proposes a transformer-based multitask model for continuous sign language recognition (CSLR) and text translation, significantly improving translation accuracy.

- Deeksha Varshney*, **Aizan Zafar***, Niranshu Behera, Asif Ekbal. "CDialog: A Multi-turn Covid-19 Conversation Dataset for Entity-Aware Dialog Generation." In Proceedings of **EMNLP 2022**, pp. 11373–11385, Abu Dhabi, UAE. (**CORE Rank: A***)
Introduces CDialog, a multi-turn COVID-19 dialogue dataset with medical entity annotations, improving patient–chatbot interactions.
- **Aizan Zafar**, K. Swarupa Rani. "Novel Initialization Strategy for K-Modes Clustering Algorithm." In Proceedings of **BigDML 2019**.
Proposes a density-based initialization method for K-modes clustering, enhancing clustering accuracy for categorical datasets.

Journal Articles

- **Aizan Zafar**, Deeksha Varshney, Sovan Kumar Sahoo, Amitava Das, Asif Ekbal. "Are My Answers Medically Accurate? Exploiting Medical Knowledge Graphs for Medical Question Answering." **Applied Intelligence**, Vol. 54, 2024. Springer. (**Impact Factor: 5.3**)
- **Aizan Zafar**, Sovan Kumar Sahoo, Deeksha Varshney, Amitava Das, Asif Ekbal. "KIMedQA: Towards Building Knowledge-Enhanced Medical QA Models." **Journal of Intelligent Information Systems**, 2024. Springer. (**Impact Factor: 4.57**)
- **Aizan Zafar**, Sovan Kumar Sahoo, Harsh Bhardwaj, Amitava Das, Asif Ekbal. "KI-MAG: A Knowledge-Infused Abstractive Question Answering System in the Medical Domain." **Neurocomputing**, Vol. 571, 2024. Elsevier. (**Impact Factor: 6.0**)
- Deeksha Varshney, **Aizan Zafar**, Niranshu Kumar Behera, Asif Ekbal. "Knowledge Graph Assisted End-to-End Medical Dialogue Generation." **Artificial Intelligence in Medicine**, Vol. 139, 2023. Elsevier. (**Impact Factor: 7.011**)
- Deeksha Varshney, **Aizan Zafar**, Niranshu Kumar Behera, Asif Ekbal. "Knowledge Grounded Medical Dialogue Generation Using Augmented Graphs." **Scientific Reports**, Vol. 13, 3310 (2023). Nature Publishing Group. (**Impact Factor: 4.996**)
- **Aizan Zafar**, Rizwan Qureshi, Mubarak Shah. "CRANE: Leveraging Visual Language Understanding for Causal Reasoning in Medical Visual Question Answering." **Transactions on Medical Imaging**, under review.
- Shaina Raza, Rizwan Qureshi, Anam Zahid, Joseph Fiorese, Ferhat Sadak, Muhammad Saeed, Ranjan Sapkota, Aditya Jain, Anas Zafar, Muneeb Ul Hassan, **Aizan Zafar**, Hasan Maqbool, Ashmal Vayani, Jia Wu, Maged Shoman. "Who is Responsible? Data, Models, or Regulations — A Comprehensive Survey on Responsible Generative AI for a Sustainable Future." **arXiv**, 2025.

Skills

Programming Languages	Python, C
Deep Learning Frameworks & Libraries	PyTorch, TensorFlow, Hugging Face Transformers, Graph Neural Networks (GNNs), Convolutional Neural Networks (CNNs)
Foundation Models	LLMs (GPT, LLaMA, BioBERT, ClinicalBERT), VLMs (BLIP, Qwen2.5)
NLP & Multimodal Skills	Question Answering, Dialogue Generation, Multimodal Reasoning, Instruction Tuning, Visual Grounding
Knowledge Graphs & Neuro-Symbolic AI	Ontology Design, Logical Rule Integration, Medical Knowledge Graphs (UMLS), PLC/SCL Knowledge Graphs (SCLKG)

Skills (continued)

Causal & Explainable AI	■ Causal Inference, Counterfactual Reasoning, Chain-of-Thought (CoT), Interpretable AI Systems
Academic & Leadership	■ Research, Teaching, Student Mentoring, Workshop Facilitation, Proposal Writing, Team Collaboration

Honors and Awards

2026	■ Paper Acceptance, CVPR 2026 Workshop (KnowledgeMR) Accepted paper " <i>Towards Responsible Multimodal Medical Reasoning via Context-Aligned Vision-Language Models.</i> "
	■ Paper Acceptance, WACV 2026 (CORE A) Accepted paper " <i>GAEA: A Geolocation Aware Conversational Assistant.</i> "
2025	■ Paper Acceptance, COLING 2025 (CORE A) Accepted paper " <i>MedEx: Enhancing Medical Question Answering with First-Order Logic Based Reasoning and Knowledge Injection.</i> "
2024	■ Paper Acceptance, EMNLP 2024 Findings (CORE A*) Accepted paper " <i>MedLogic-AQA: Enhancing Medical Question Answering with Abstractive Models Focusing on Logical Structures.</i> "
2024–Present	■ Peer Reviewer, Top-tier AI Venues Reviewer for ACL ARR, EMNLP, CVPR, WACV, IEEE JBHI, and leading AI conferences/journals.
2019	■ Ph.D. Fellowship (GATE), Government of India National-level merit fellowship based on GATE (Computer Science), funded by MHRD/AICTE.
2017	■ M.Tech Fellowship (GATE), Government of India Merit-based postgraduate scholarship awarded via national GATE ranking.

References

1. **Prof. Mubarak Shah** (FIEEE, FIAPR, Director, CRCV)
University of Central Florida, Orlando, USA
✉ shah@crcv.ucf.edu
🌐 <https://crcv.ucf.edu/person/mubarak-shah/>

2. **Associate Prof. Asif Ekbal**
Department of Computer Science and Engineering
Indian Institute of Technology Patna, India
✉ asif.ekbal@gmail.com; asif@iitp.ac.in
🌐 <https://ekbalasif.github.io/>